HP_power_board

File: HP_power_board.kicad_sch

LP_logic_board

File: LP_logic_board.kicad_sch

pi_power_stub

File: pi_power_stub.kicad_sch

pack_sense_board

File: pack_sense_board.kicad_sch

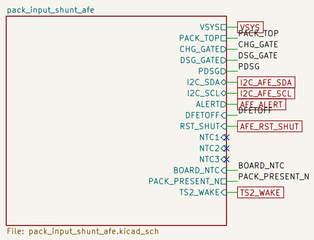
marengo

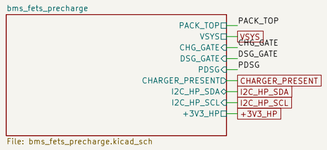
Sheet: /

File: marengo_pdu.kicad_sch

Title: marengo power board rev A

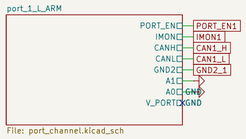
Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 1/27

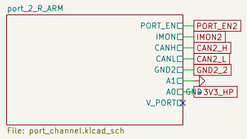


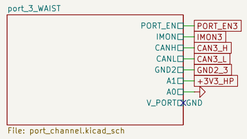


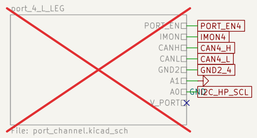


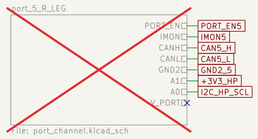


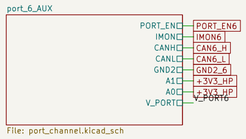




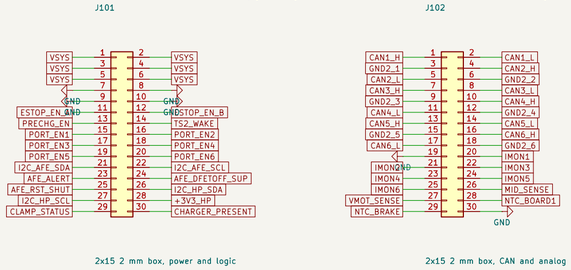


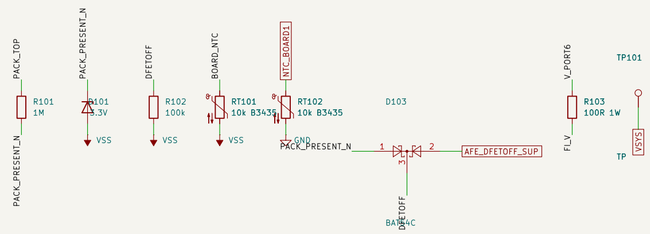






Board-to-board: two 2x15 2 mm ribbons to LP, pin for pin. No stock 2x30 latched header exists. All 60 pins used.





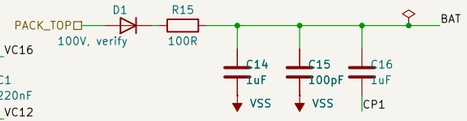
DFETOFF, active high (config 0x02): pack sense harness pulled, or supervisor backup, opens DSG. E-stop loop deliberately not wired here: it would deadlock boot (12 V comes after DSG) and drop compute.
Pull-up from PACK_TOP, not LP: it holds with the LP rails down. 54 uA through the presence loop when mated.
Fault injection: test mode, disarmed. Short H-L, H-GND2, L-GND2, H-VBUS via 100R.

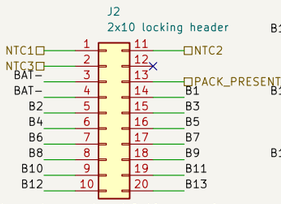
marengo
Sheet: /HP_power_board/
File: HP_power_board.kicad_sch

Title: HP_power_board: high-power board

Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 2/27

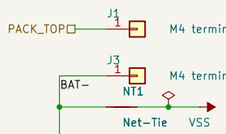
BAT from the pack + terminal, so the AFE stays up if J2 is pulled
with pack power still connected. Fig 16-2: diode, 100R, 1uF, 100pF. CP1 is 1uF, not 0.47uF: see sheet bms_fets_precharge. Caps 100 V.



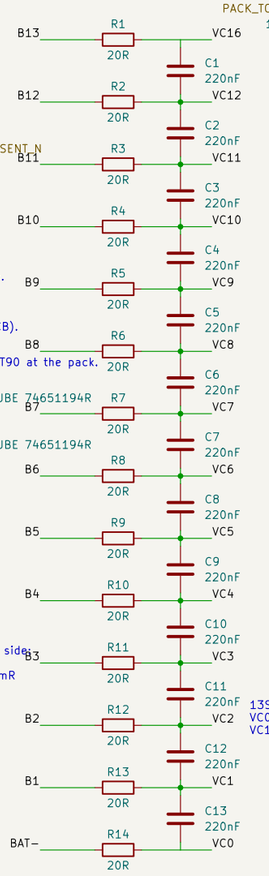


J2: harness to the pack sense board 2x10, same pin map.
Columns 4-10 zigzag B0..B13: a bent pin shorts 1 cell
across a column, 2 cells along a row.
NTC return = B0 at the pack sense board. Spare: pin 12.
B0 lands on BAT- (datasheet p70: VC0 tied to VSS on PCB).

J1 pack +, J3 pack -: 10 AWG ring-lug pigtails to the XT90 at the pack.

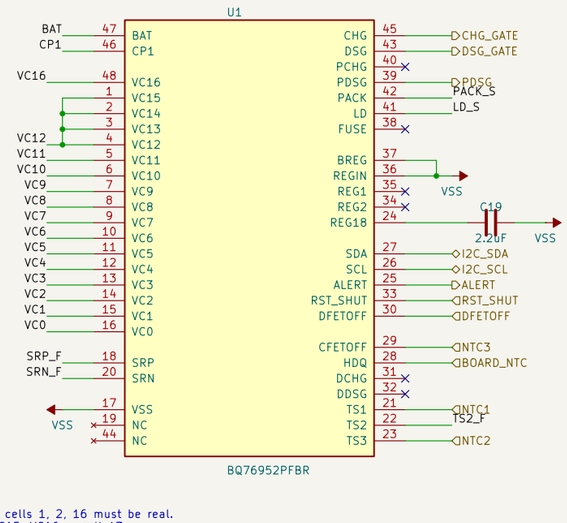


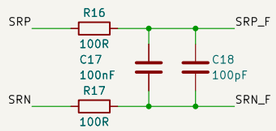
Pack negative -> shunt -> system GND. SRP is the cell side;
charge current makes SRP positive (datasheet p5).
AFE VSS ties at BAT-, so VSS sits above GND by I x 0.3 mR
in discharge: 60 mV at the 200 A SCD point.



13S per Table 16-3 (p75): cells 1, 2, 16 must be real.
VC0..VC12 = cells 1-12, VC15-VC16 = cell 13.
VC13-15 short to VC12. Cell inputs in use: 1-12 and 16.

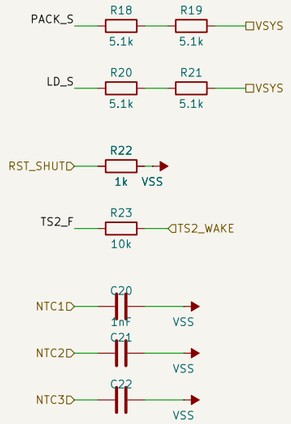
Cell filter: 20R + 220nF on every used input
(datasheet p9: RC 20-100R, CC 0.1/0.22/1 uF; p67 Fig 16-2).





p65, p76: 100R each side, 100nF + 100pF differential, near U1,
symmetric, same PCB side. Kelvin SRP/SRN routed in parallel from RS1.

PACK and LD both sense VSYS, the common port after the FETs
(Fig 16-3 R42-R45). PCHG, FUSE, DCHG, DDSG unused, left floating (p75).



Four thermistors: TS1, TS3, CFETOFF (NTC1-3), HDQ (board).
TS2 stays free as SHUTDOWN wake (p22 pin table, section 13):
switch to VSS through 10k (Fig 16-2 R27). 1nF: p65 limit <4nF.
REGIN, BREG tied to VSS, REG1, REG2 open (p75): I2C pull-ups on LP.
CFETOFF as NTC needs DFETOFF set as BOTHOFF to stop CHG by pin (p48).

SIZING (shunt 0.3 mOhm: no 0.25 mOhm 4-terminal part exists; datasheet p22 steps)
OCD1 20 mV = 66.7 A (hardware comparator, up to 425 ms) OCD3 60 A for 1 s (current-based, inside the AFE, the agreed 60 A ceiling)
OCC 14 mV = 46.7 A at 425 ms SCD 60 mV = 200 A, +/-15% = 170-230 A, delay 15-450 us in 15 us steps (keeps the 150 A, 100 ms path rating)
OCD and OCC are set in 2 mV steps. Accuracy below 20 mV is -2/+2.65 mV (p22), about -6.7/+8.8 A on this shunt.

OPEN ON THIS SHEET
1. External balance FETs (PLAN 3.1, 150 mA per cell) are not captured. They change the cell RC network: revisit 20R.
2. CP1 raised to 1 uF for the 4+4 FET gate load (p65 rule): worst case 1.69 uC droops 11 V by 1.7 V, pump start about 130 ms.
3. OTP owed: DFETOFF active high (0x02), CFETOFF and HDQ as thermistor inputs. Presence and DFETOFF logic are on HP_power_board.
4. RS1 land: CSS4J-4026 on the BVR4026 library footprint, check against the Bourns drawing.

Pack terminals, sense header, shunt, BQ76952 AFE. Part numbers verify. TI SLUSE13B.

marengo

Sheet: /HP_power_board/pack_input_shunt_afe/
File: pack_input_shunt_afe.kicad_sch

Title: pack_input_shunt_afe

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 3/27

PDSG FET bypasses DSG only (Fig 16-1): source at CD. Driver sinks 30 uA, so gate-source >= 1M (p45).

PDSG
R210 1M
Q209 DMP10H400SE
R209 100R

Common drain (CD): Fig 16-3 pin numbering, gates referenced to the outer sources. Body diodes oppose each other.

CD
Q201 Q202 Q203 Q204 Q205 Q206 Q207 Q208
CHG_G R201 4.7R
CHG_G R202 4.7R
CHG_G R203 4.7R
CHG_G R204 4.7R
DSG_G R205 4.7R
DSG_G R206 4.7R
DSG_G R207 4.7R
DSG_G R208 4.7R
IPT015N10N5
PACK_TOP
DSG_OUT
RS201 0.3mR CSS4J-4026 5W
VSYS_SENS_P
VSYS_SENS_N
VSYS
TP201 VSYS
D207 SMDJ58A GND
C204 100nF 100V
C205 100nF 100V 100uF 80V placeholder
C206 GND

CHG bank: sources on the stack

DSG bank: sources on the output

CHG drive, Fig 16-3 R37/R31/D5.

CHG_GATED
R211 5.1k
R212 10M
D201 MMSZ5246B 16V
PACK_TOP
CHG_G

Fig 16-3 C28 C29: across the FET, two in series.

C201 100nF 100V
C202 100nF 100V

Stack clamp, added: inductive spike at the top of stack at fast turn-off (p72). Takes 0.36 J at 2 uH, 600 A.

D206 SMDJ58A GND

Fig 16-3 Q4 D7 R33 R35 D8: If the output is yanked below GND, Q210 holds the DSG gate at its source.

DSG drive. On: 10.2k (Fig 16-3 R39+R41). Off: 220R + diode, NOT the 7.5k of Fig 16-3 (see note 2).

DSG_GATED
R213 5.1k
R214 5.1k
D202 BAV19W
R215 220R
R216 10M
D203 MMSZ5246B 16V
R218 10k
D204 MMSZ5246B 16V
D205 BAV19W
R217 10M
Q210 BSS123
DSG_G
DSG_OUT

U201 INA228AIDGSR
+3V3_HP
VSYS
VSYS_SENS_P
VSYS_SENS_N
C203 100nF
I2C_HP_SCL
I2C_HP_SDA
GND

Main path meter, PLAN 3.9. Address 0x40: A1, A0 to GND. +/-40.96 mV range = +/-164 A.

Charge input, common port: the charger joins on VSYS, DSG body diode side, as the BQ76952 expects (PACK pin sees it, p45). Also wakes the AFE from ship mode through LD.
PLAN 3.7 says reverse-polarity FET. With a reversed charger the blocker sees 54 + 54 = 108 V: above the 100 V FET and the LM74700 (75 V, SNOSD17G). So a 200 V Schottky: 0.7 V drop, 3 W at 4 A. See RATIONALE-log.
CHARGER_PRESENT: CHG_IN only, so VSYS cannot fake it: high above 34 V, 3.2 V at 54.6 V.

J201
CHG_IN
D220 V15PM20 200V 15A
VSYS
D208 SMDJ58CA
D209 BAV19W
R219 1M
R220 62k
D210 MMSZ4685 3.6V
C207 1nF
CHARGER_PRESENT
J202
GND

SIZING (design/README 2.2)
Conduction, 80 A: 2 x 0.6 mOhm hot x 80^2 = 7.7 W total. At 10 to 30 A: 0.1 to 1.1 W for these two banks.
FET: IPT015N10N5 Rds 1.5 mOhm max at Vgs 10 V, Vgs +/-20 V, so 11 V drive and 16 V zeners are in range (p69 needs >= 15 V).
Shunt voltage rule p69: 13S x 5 V = 65 V DC, 130 V peak. 100 V FETs meet DC. Peak: both TVS parts hold it under 100 V.

SOA must cover 600 A prospective for about 50 us, check at gate. Worked here, gate review to confirm:
1. Prospective current: 600 A / 4 = 150 A per FET. On-state part: 0.6 mOhm x 600 A = 0.36 V, 216 W bank, nothing.
2. Turn-off: TI's 4.3k path scaled to 4 x 12 nF has a Miller time of about 120 us at full bus: 100 us SOA is about 45 A per FET at 54 V. FAILS.
   With 220R: gate 48 nF, tau 10.6 us, plateau reached in 8 us, Miller 136 nC / 23 mA = 6 us (9 us max Qgd). 10 us SOA at 54 V is about 300 A per FET. PASSES at 150 A.
   Peak dT = 8.2 kW x 0.009 K/W (Zth 10 us) = 74 K. Margin shrinks with poor sharing at turn-off: scope it at bring-up.
3. Spike: p72 warns fast turn-off raises the top of stack. D206 clamps it. Tune 220R upward on the bench if the clamp runs hot.

CP1 is 1 uF on sheet 1: 8 x 211 nC max = 1.69 uC at p65's rule, droop 1.7 V from 11 V.
PDSG not PCHG: p45 PDSG charges the LOAD before DSG closes; PCHG charges a LOW BATTERY. README 2.1 described the PDSG job.
100R: peak 54.6^2/100 = 298 W. Stored energy 0.5 x 1.1 mF x 54.6^2 = 1.6 J per power-up, tau 0.11 s. Part must be pulse rated.
Set OTP: PREDISCHARGE on, timeout > 0.6 s (5 tau).

Fig 16-1, Fig 16-3 of TI SLUSE13B. High side, common drain, common port.

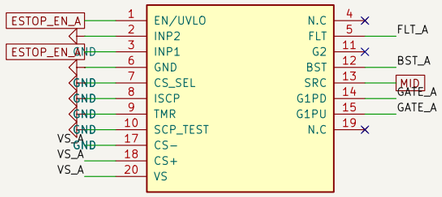
marengo

Sheet: /HP_power_board/bms_fets_precharge/
File: bms_fets_precharge.kicad_sch

Title: bms_fets_precharge: BMS CHG/DSG banks, predischarge, charge input

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 4/27

U301



TPS48100QDGXRQ1

R301 100R

C301 100nF 100V

C302 1uF 25V X7R

R302 100k

R303 100k

R304 1M

TP301

TP

FLT_A

VS_A

GND

GND

Bank A: VSYS to MID

Drain toward VSYS, source toward VMOT: body diodes conduct VMOT to VSYS, so regen still reaches the pack with the banks open (PLAN 3.4).

Q301

Q303

Q305

IPT015N10N5

IPT015N10N5

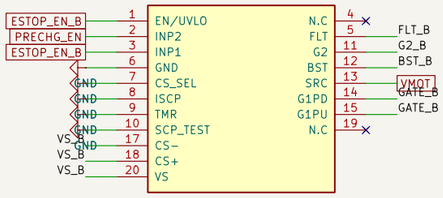
IPT015N10N5

R309 2.2R

R310 2.2R

R311 2.2R

U302



TPS48100QDGXRQ1

R305 100R

C303 100nF 100V

C304 1uF 25V X7R

R306 100k

R307 100k

R308 1M

TP302

TP

FLT_B

VS_B

GND

GND

Bank B: MID to VMOT

Q302

Q304

Q306

IPT015N10N5

IPT015N10N5

IPT015N10N5

R312 2.2R

R313 2.2R

R314 2.2R

VMOT precharge (new: IPT015N10N5 SOA is 0.7 A at 54 V, 10 ms, so a hard close on 1.8 mF is out). Close A, MID rises. PRECHG_EN high drives G2. Wait VMOT within 5 % of MID, then close B. EN low kills G2 too, so the loop gates precharge in hardware. Energy 2.68 J at 1.8 mF, tau 180 ms, peak 29.8 W.

Status (TPS48100 has no VDS output): MID, VMOT dividers 1M / 56k, 3.3 V = 62 V, plus VSYS on LP. Bank A short reads MID near VSYS with both open and VMOT low. FLT covers short circuit, charge pump UV, UVLO and test point, not routed to LP.

Q307

R315 100R PWR263S-3108-F

R316 100R PWR263S-3108-F

R317 1M

C305 100nF 100V

R318 1M

R319 56k

C306 1nF

R320 49.9R

R321 1M

R322 56k

C307 1nF

R323 49.9R

TP303

TP304

TP305

TP

TP

TP

GND

GND

GND

GND

GND

GND

TPS48100−Q1 simple−driver mode (SLVSGC0 Fig 7−11). No firmware in the enable path.

marengo

Sheet: /HP_power_board/estop_banks/

File: estop_banks.kicad_sch

Title: estop_banks: E−stop banks A and B, VMOT precharge

Size: A3	Date: 2026−11−06	**Rev: A−wip**
KiCad E.D.A. 10.0.8		Id: 5/27

Bulk: 18 x 120 uF 80 V hybrid, 1.8 mF. 220 uF 80 V 10x12.5 does not exist; can is 10 x 16.5 mm; footprint is the closest library match, real land owed.

C401 100uF 80V
C402 100uF 80V
C403 100uF 80V
C404 100uF 80V
C405 100uF 80V
C406 100uF 80V
C407 100uF 80V
C408 100uF 80V
C409 100uF 80V
C410 100uF 80V
C411 100uF 80V
C412 100uF 80V
C413 100uF 80V
C414 100uF 80V
C415 100uF 80V
C416 100uF 80V
C417 100uF 80V
C418 100uF 80V
D401 SMDJ58A-GRA101MJH0G

D402 SMDJ58A

C419 1uF 100V
C420 4.7uF 25V
R401 464k
R402 49.9k
R403 9.09k

U401 TPS7A4001DGNR

U403 LM4040AIM3-2.5

Clamp: 57.50 V on, 56.50 V off (VOH 11.7 V, 19.1M hysteresis). Spread 56.82–58.19 V full temperature, 56.99–58.02 V 0–60 C. 10 Ohm 50 W clad external: 330 W, 5.75 A.
INA228 0x4D: clamp power and energy on its own accumulator. Budget about 5 mA, 0.23 W in the LDO.
Failure: stuck-on 330 W caught by brake NTC and opening the E-stop banks. SMDJ58A stays off under 64.4 V, spike protection only.
OPEN (Joey): 30 A regen cap against 5.75 A clamp. BMS opening mid-brake rises 13.5 V/ms, drives stop at 60 V in about 185 us. Kept 10 Ohm: a drive OV stop is safe; 2 Ohm means 1.7 kW peak.

R404 499k 0.1%
R405 619k 0.1%
R406 51.1k 0.1%
R407 19.1M
C421 100pF C0G
C422 100nF

U402 TLV1805DBVR

Q402 BSS123

J401
J402 M4 terminal, brake R +
J403 brake R NTC
M4 terminal, brake R –

C423 1uF 25V
R408 4.7R
R409 100k

U404 UCC27517DBVR

Q401 IPT015N10N5

RS401 20mR 4T 1W

D403 MBRS3100T3G
R410 10R
R411 10R
C424 100nF
R412 100R
C425 100nF

U405 INA228AIDGSR

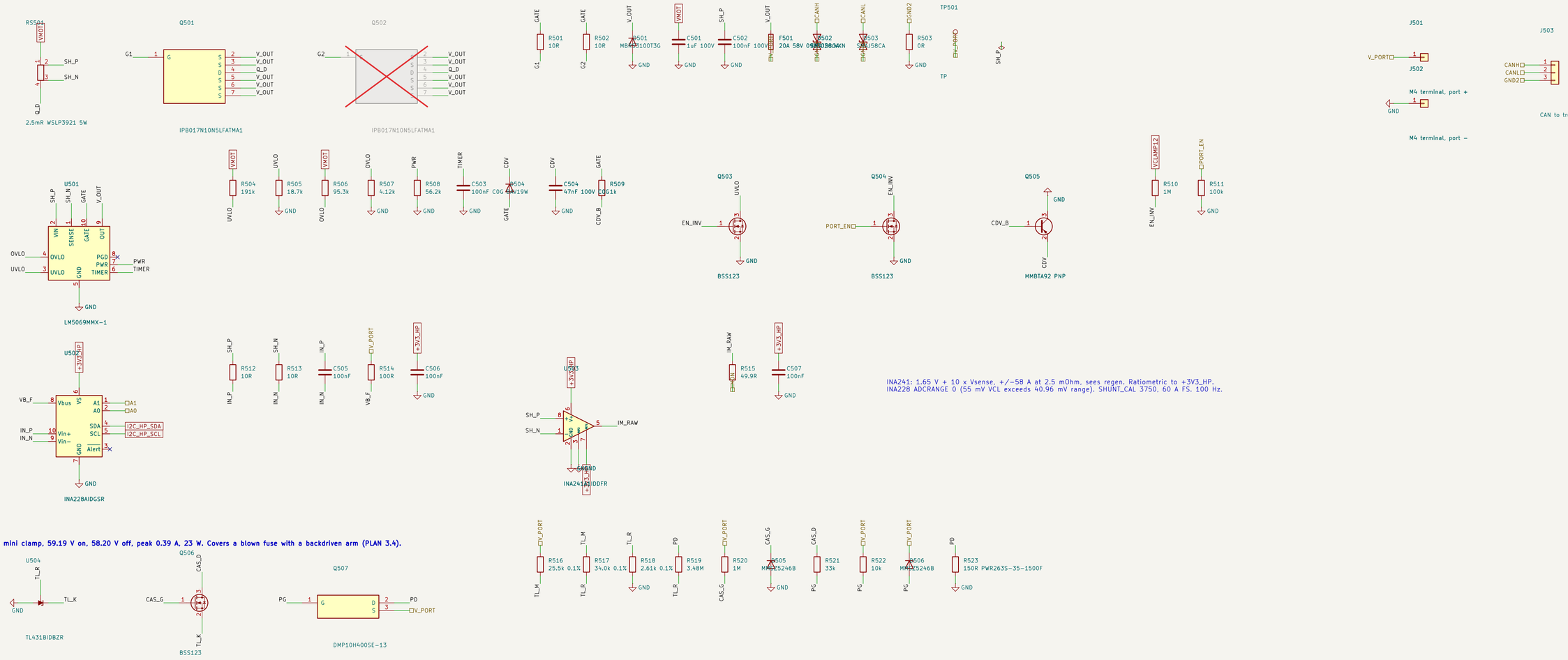
Regen clamp powers itself from VMOT, works with supervisor, BMS and pack all off (PLAN 3.4).

marengo

Sheet: /HP_power_board/vmot_bulk_clamp/
File: vmot_bulk_clamp.kicad_sch

Title: vmot_bulk_clamp: VMOT bulk, TVS, self-powered regen clamp

Size: A2	Date: 2026-11-06	Rev: A-wip
KiCad E.D.A. 10.0.8		Id: 6/27



Load-side mini clamp, 59.19 V on, 58.20 V off, peak 0.39 A, 23 W. Covers a blown fuse with a backdriven arm (PLAN 3.4).

Values shown are the 22 A arm, waist, aux channel. Leg channels (DNP in rev A) change: Q502 fitted; RS501 1 mOhm (48.5–61.5 A) or 1.5 mOhm (32–41 A), no 1.2 mOhm part exists; RPWR 23.7k at 1 mOhm, 30.1k at 1.2 mOhm; INA228 SHUNT_CAL rescaled; fuse needs another format (997 MINI stops at 30 A).
Enable: PORT_EN high (3V3, 100k down) turns Q3 on, Q2 off, UVLO released. Supervisor dead: Q2 on, UVLO low, port off. Latch reset: drop PORT_EN, wait 148 ms, raise it.
ILIM 19.4–24.6 A, breaker 32–52 A, fault timer 2.8–9.0 ms, UVLO on 32.0 V off 28.0 V, OVLO off 60.3 V. Starts on dV/dt: 47 nF covers 3.6 mF of drive capacitance.
GND2: bus reference ties to the port return here. Galvanic isolation then separates LP logic ground from motor return; it does not block a CAN-to-VBUS fault, ISOW1044 survives that at +/−58 V.
SMAJ58CA for sustained 54 V survival (NUP2105L VBR 26–32 V fails). Its clamp is above the ISOW1044 58 V limit and its capacitance is unchecked: scope on the bus.

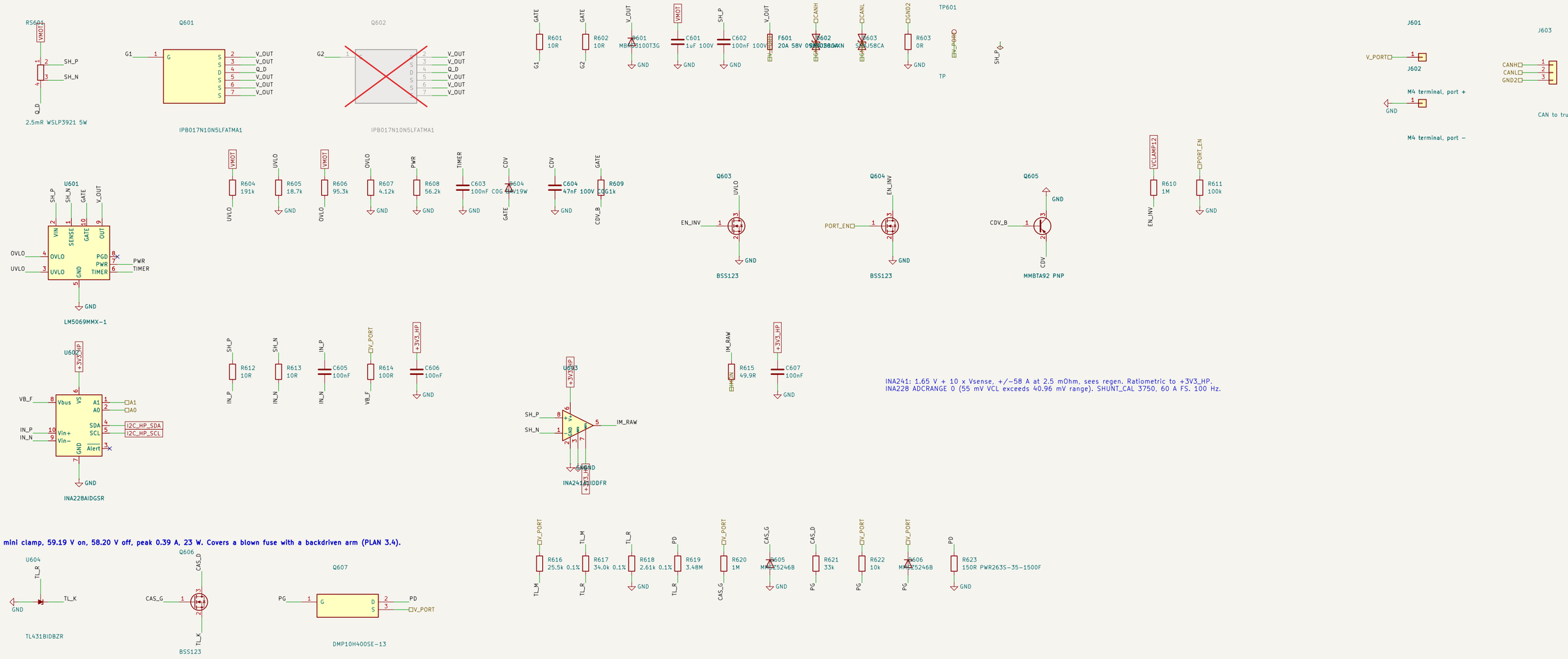
LM5069-1 latched hot swap, INA228 + INA241 on one shunt, blade fuse, load-side mini clamp.

marengo

Sheet: /HP_power_board/port_1_L_ARM/
File: port_channel.kicad_sch

Title: port_channel: one limb port (instanced x6, legs DNP)

Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 7/27



Load-side mini clamp, 59.19 V on, 58.20 V off, peak 0.39 A, 23 W. Covers a blown fuse with a backdriven arm (PLAN 3.4).

Values shown are the 22 A arm, waist, aux channel. Leg channels (DNP in rev A) change: Q502 fitted; RS501 1 mOhm (48.5–61.5 A) or 1.5 mOhm (32–41 A), no 1.2 mOhm part exists; RPWR 23.7k at 1 mOhm, 30.1k at 1.2 mOhm; INA228 SHUNT_CAL rescaled; fuse needs another format (997 MINI stops at 30 A).
Enable: PORT_EN high (3V3, 100k down) turns Q3 on, Q2 off, UVLO released. Supervisor dead: Q2 on, UVLO low, port off. Latch reset: drop PORT_EN, wait 148 ms, raise it.
ILIM 19.4–24.6 A, breaker 32–52 A, fault timer 2.8–9.0 ms, UVLO on 32.0 V off 28.0 V, OVLO off 60.3 V. Starts on dV/dt: 47 nF covers 3.6 mF of drive capacitance.
GND2: bus reference ties to the port return here. Galvanic isolation then separates LP logic ground from motor return: it does not block a CAN-to-VBUS fault, ISOW1044 survives that at +/−58 V.
SMAJ58CA for sustained 54 V survival (NUP2105L VBR 26–32 V fails). Its clamp is above the ISOW1044 58 V limit and its capacitance is unchecked: scope on the bus.

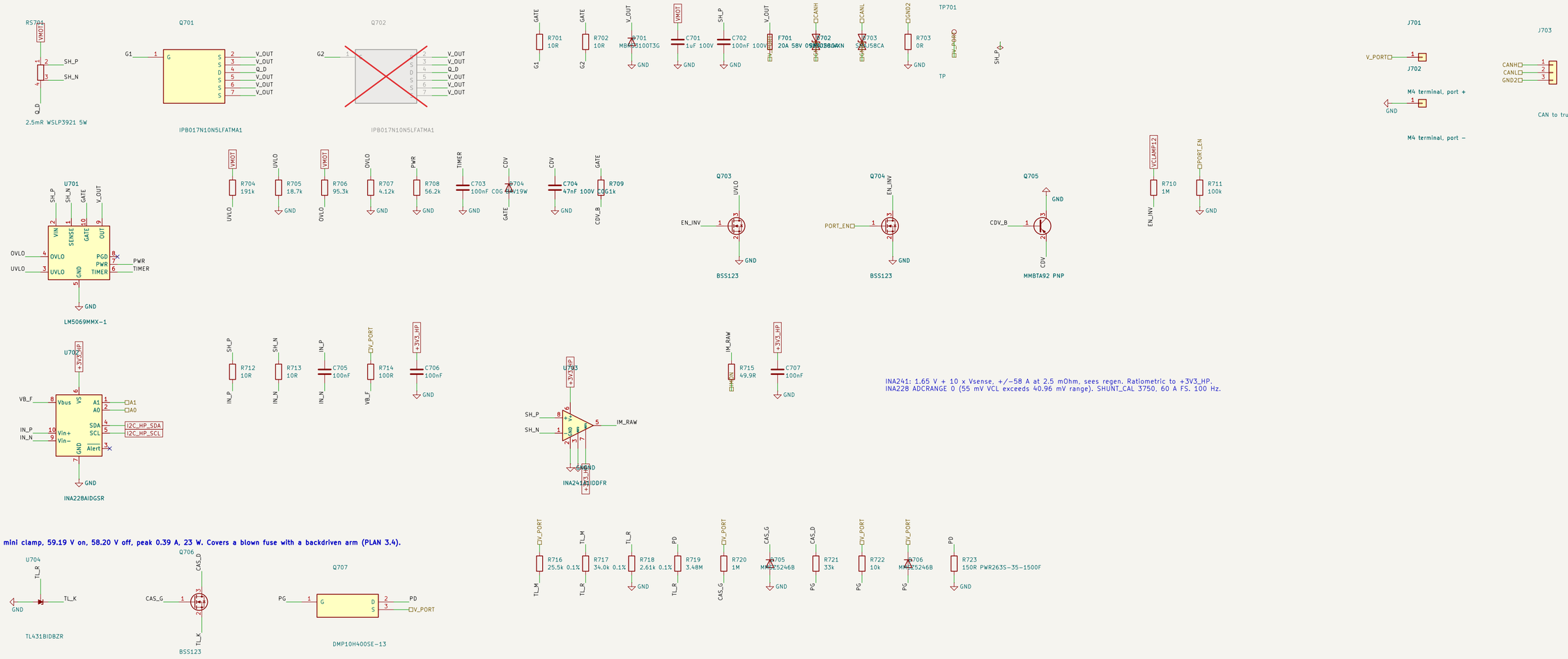
LM5069-1 latched hot swap, INA228 + INA241 on one shunt, blade fuse, load-side mini clamp.

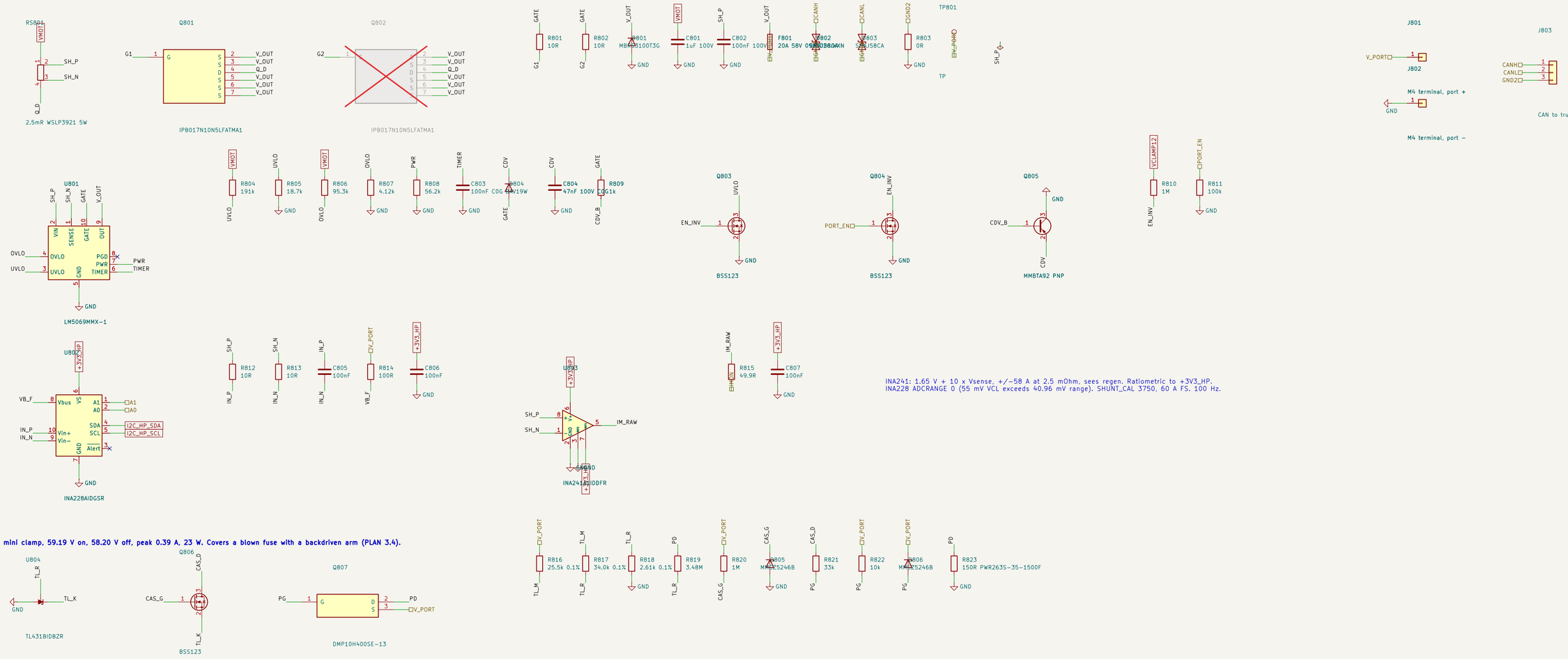
marengo

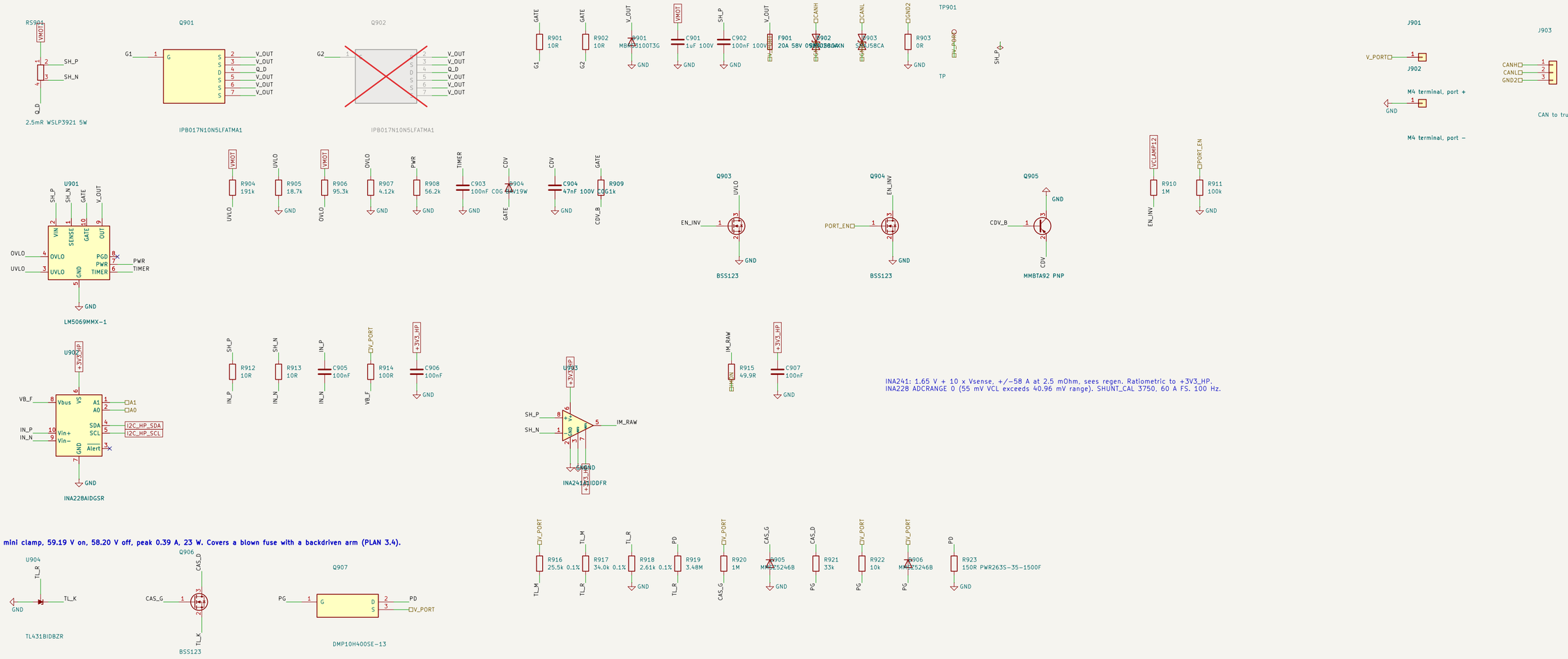
Sheet: /HP_power_board/port_2_R_ARM/
File: port_channel.kicad_sch

Title: port_channel: one limb port (instanced x6, legs DNP)

Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 8/27


RS701
VMOT
SH_P
SH_N
Q_D
2.5mR WSLP3921 5W
Q701
G1
V_OUT
Q_D
IPB017N10N5LFATMA1
Q702
G2
IPB017N10N5LFATMA1
GATE
R701 10R
R702 10R
V_OUT
D701 MBR20100T3G
C701 1uF 100V
C702 100nF 100V
SH_P
F701 20A 58V 0R
CANH
CANL
GND2
R703 0R
TP701
TP
J701
V_PORT
J702
M4 terminal, port +
M4 terminal, port −
J703
CANH
CANL
GND2
CAN to trunk
U701
LM5069MMX-1
VIN
SENSE
GATE
OUT
OVLO
UVLO
PGD
PWR
TIMER
R704 191k
R705 18.7k
R706 95.3k
R707 4.12k
R708 56.2k
C703 100nF C0G
D704 V19W
C704 47nF 100V
R709 C0G 1k
Q703
BSS123
EN_INV
Q704
PORT_END
Q705
MMBTA92 PNP
CDV_B
VCLAMP12
R710 1M
PORT_EN
R711 100k
U702
INA228AIDGSR
+3V3_HP
VB_F
IN_P
IN_N
A1
A0
I2C_HP_SDA
I2C_HP_SCL
R712 10R
R713 10R
C705 100nF
R714 100R
C706 100nF
U703
INA241A3IDDFR
IM_RAW
R715 49.9R
C707 100nF
INA241: 1.65 V + 10 x Vsense, +/−58 A at 2.5 mOhm, sees regen. Ratiometric to +3V3_HP.
INA228 ADCRANGE 0 (55 mV VCL exceeds 40.96 mV range). SHUNT_CAL 3750, 60 A FS. 100 Hz.
Load−side mini clamp, 59.19 V on, 58.20 V off, peak 0.39 A, 23 W. Covers a blown fuse with a backdriven arm (PLAN 3.4).
U704
TL431BIDBZR
TL_R
TL_K
Q706
BSS123
CAS_D
CAS_G
Q707
DMP10H400SE-13
PG
PD
R716 25.5k 0.1%
R717 34.0k 0.1%
R718 2.61k 0.1%
R719 3.48M
R720 1M
D705 MMBZ5246B
R721 33k
R722 10k
D706 MMBZ5246B
R723 150R PWR263S-35-1500F
TL_M
TL_L
Values shown are the 22 A arm, waist, aux channel. Leg channels (DNP in rev A) change: Q502 fitted; RS501 1 mOhm (48.5−61.5 A) or 1.5 mOhm (32−41 A), no 1.2 mOhm part exists;
RPWR 23.7k at 1 mOhm, 30.1k at 1.2 mOhm; INA228 SHUNT_CAL rescaled; fuse needs another format (997 MINI stops at 30 A).
Enable: PORT_EN high (3V3, 100k down) turns Q3 on, Q2 off, UVLO released. Supervisor dead: Q2 on, UVLO low, port off. Latch reset: drop PORT_EN, wait 148 ms, raise it.
ILIM 19.4−24.6 A, breaker 32−52 A, fault timer 2.8−9.0 ms, UVLO on 32.0 V off 28.0 V, OVLO off 60.3 V. Starts on dV/dt: 47 nF covers 3.6 mF of drive capacitance.
GND2: bus reference ties to the port return here. Galvanic isolation then separates LP logic ground from motor return: it does not block a CAN−to−VBUS fault, ISOW1044 survives that at +/−58 V.
SMAJ58CA for sustained 54 V survival (NUP2105L VBR 26−32 V fails). Its clamp is above the ISOW1044 58 V limit and its capacitance is unchecked: scope on the bus.
LM5069−1 latched hot swap, INA228 + INA241 on one shunt, blade fuse, load−side mini clamp.
marengo
Sheet: /HP_power_board/port_3_WAIST/
File: port_channel.kicad_sch
Title: port_channel: one limb port (instanced x6, legs DNP)
Size: A2
Date: 2026−11−06
Rev: A−wip
KiCad E.D.A. 10.0.8
Id: 9/27


RS801
VMOT
SH_P
SH_N
Q_D
2.5mR WSLP3921 5W
Q801
G1
V_OUT
Q_D
IPB017N10N5LFATMA1
Q802
G2
IPB017N10N5LFATMA1
GATE
R801 10R
R802 10R
V_OUT
D801 MBRB2100T3G
GND
VMOT
C801 1uF 100V
C802 100nF 100V
SH_P
V_OUT
F801 20A 58V 0R2380WKN
CANH
D802 SMAJ58CA
CANL
D803 SMAJ58CA
GND2
R803 0R
TP801
TP
SH_P
J801
V_PORT
J802
M4 terminal, port +
M4 terminal, port −
J803
CANH
CANL
GND2
CAN to trunk
U801
SH_P
SH_N
GATE
V_OUT
VIN
SENSE
GATE
OUT
OVLO
UVLO
PGD
PWR
TIMER
LM5069MMX-1
R804 191k
R805 18.7k
R806 95.3k
R807 4.12k
R808 56.2k
C803 100nF C0G
D804 CV19W
C804 47nF 100V
R809 C0G1k
CDV_B
Q803
EN_INV
BSS123
Q804
PORT_END
Q805
MMBTA92 PNP
VCLAMP12
R810 1M
R811 100k
PORT_EN
U802
+3V3_HP
Vbus
VS
A1
A0
SDA
SCL
Vin+
Vin−
Alert
I2C_HP_SDA
I2C_HP_SCL
VB_F
IN_P
IN_N
INA228AIDGSR
R812 10R
R813 10R
C805 100nF
R814 100R
C806 100nF
CV_PORT
U803
IM_RAW
INA241A3IDDFR
R815 49.9R
C807 100nF
INA241: 1.65 V + 10 x Vsense, +/−58 A at 2.5 mOhm, sees regen. Ratiometric to +3V3_HP.
INA228 ADCRANGE 0 (55 mV VCL exceeds 40.96 mV range). SHUNT_CAL 3750, 60 A FS. 100 Hz.
Load−side mini clamp, 59.19 V on, 58.20 V off, peak 0.39 A, 23 W. Covers a blown fuse with a backdriven arm (PLAN 3.4).
U804
TL_R
TL_K
GND
TL431BIDBZR
Q806
CAS_D
CAS_G
TL_K
BSS123
Q807
PG
PD
V_PORT
DMP10H400SE−13
TL_M
TL_R
R816 25.5k 0.1%
R817 34.0k 0.1%
R818 2.61k 0.1%
R819 3.48M
R820 1M
D805 MMBZ5246B
R821 33k
R822 10k
D806 MMBZ5246B
R823 150R PWR263S−35−1500F
Values shown are the 22 A arm, waist, aux channel. Leg channels (DNP in rev A) change: Q502 fitted; RS501 1 mOhm (48.5−61.5 A) or 1.5 mOhm (32−41 A), no 1.2 mOhm part exists;
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Enable: PORT_EN high (3V3, 100k down) turns Q3 on, Q2 off, UVLO released. Supervisor dead: Q2 on, UVLO low, port off. Latch reset: drop PORT_EN, wait 148 ms, raise it.
ILIM 19.4−24.6 A, breaker 32−52 A, fault timer 2.8−9.0 ms, UVLO on 32.0 V off 28.0 V, OVLO off 60.3 V. Starts on dV/dt: 47 nF covers 3.6 mF of drive capacitance.
GND2: bus reference ties to the port return here. Galvanic isolation then separates LP logic ground from motor return: it does not block a CAN−to−VBUS fault, ISOW1044 survives that at +/−58 V.
SMAJ58CA for sustained 54 V survival (NUP2105L VBR 26−32 V fails). Its clamp is above the ISOW1044 58 V limit and its capacitance is unchecked: scope on the bus.
LM5069−1 latched hot swap, INA228 + INA241 on one shunt, blade fuse, load−side mini clamp.
marengo
Sheet: /HP_power_board/port_4_L_LEG/
File: port_channel.kicad_sch
Title: port_channel: one limb port (instanced x6, legs DNP)
Size: A2
Date: 2026−11−06
Rev: A−wip
KiCad E.D.A. 10.0.8
Id: 10/27



Load-side mini clamp, 59.19 V on, 58.20 V off, peak 0.39 A, 23 W. Covers a blown fuse with a backdriven arm (PLAN 3.4).

Values shown are the 22 A arm, waist, aux channel. Leg channels (DNP in rev A) change: Q502 fitted; RS501 1 mOhm (48.5–61.5 A) or 1.5 mOhm (32–41 A), no 1.2 mOhm part exists; RPWR 23.7k at 1 mOhm, 30.1k at 1.2 mOhm; INA228 SHUNT_CAL rescaled; fuse needs another format (997 MINI stops at 30 A).
Enable: PORT_EN high (3V3, 100k down) turns Q3 on, Q2 off, UVLO released. Supervisor dead: Q2 on, UVLO low, port off. Latch reset: drop PORT_EN, wait 148 ms, raise it.
ILIM 19.4–24.6 A, breaker 32–52 A, fault timer 2.8–9.0 ms, UVLO on 32.0 V off 28.0 V, OVLO off 60.3 V. Starts on dV/dt: 47 nF covers 3.6 mF of drive capacitance.
GND2: bus reference ties to the port return here. Galvanic isolation then separates LP logic ground from motor return; it does not block a CAN-to-VBUS fault, ISOW1044 survives that at +/−58 V.
SMAJ58CA for sustained 54 V survival (NUP2105L VBR 26–32 V fails). Its clamp is above the ISOW1044 58 V limit and its capacitance is unchecked: scope on the bus.

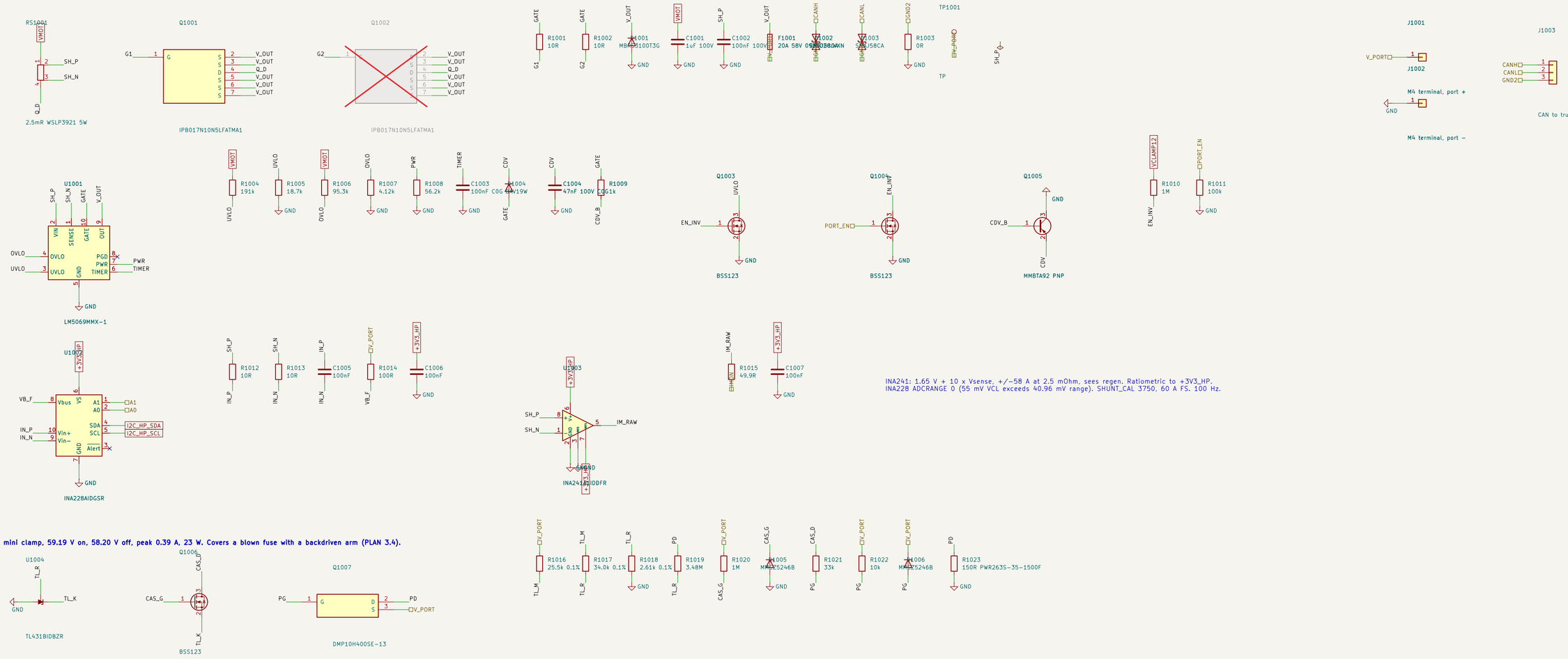
LM5069-1 latched hot swap, INA228 + INA241 on one shunt, blade fuse, load-side mini clamp.

marengo

Sheet: /HP_power_board/port_5_R_LEG/
File: port_channel.kicad_sch

Title: port_channel: one limb port (instanced x6, legs DNP)

Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 11/27



Values shown are the 22 A arm, waist, aux channel. Leg channels (DNP in rev A) change: Q502 fitted; RS501 1 mOhm (48.5−61.5 A) or 1.5 mOhm (32−41 A), no 1.2 mOhm part exists; RPWR 23.7k at 1 mOhm, 30.1k at 1.2 mOhm; INA228 SHUNT_CAL rescaled; fuse needs another format (997 MINI stops at 30 A).
Enable: PORT_EN high (3V3, 100k down) turns Q3 on, Q2 off, UVLO released. Supervisor dead: Q2 on, UVLO low, port off. Latch reset: drop PORT_EN, wait 148 ms, raise it.
ILIM 19.4−24.6 A, breaker 32−52 A, fault timer 2.8−9.0 ms, UVLO on 32.0 V off 28.0 V, OVLO off 60.3 V. Starts on dV/dt: 47 nF covers 3.6 mF of drive capacitance.
GND2: bus reference ties to the port return here. Galvanic isolation then separates LP logic ground from motor return; it does not block a CAN−to−VBUS fault, ISOW1044 survives that at +/−58 V.
SMAJ58CA for sustained 54 V survival (NUP2105L VBR 26−32 V fails). Its clamp is above the ISOW1044 58 V limit and its capacitance is unchecked: scope on the bus.

LM5069−1 latched hot swap, INA228 + INA241 on one shunt, blade fuse, load−side mini clamp.

marengo

Sheet: /HP_power_board/port_6_AUX/
File: port_channel.kicad_sch

Title: port_channel: one limb port (instanced x6, legs DNP)

Size: A2	Date: 2026−11−06	**Rev: A−wip**
KiCad E.D.A. 10.0.8		Id: 12/27

















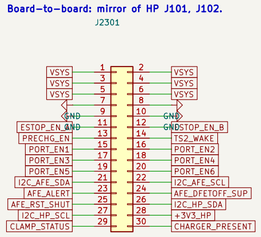


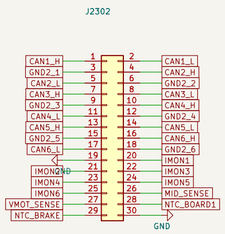






Board-to-board: mirror of HP J101, J102.



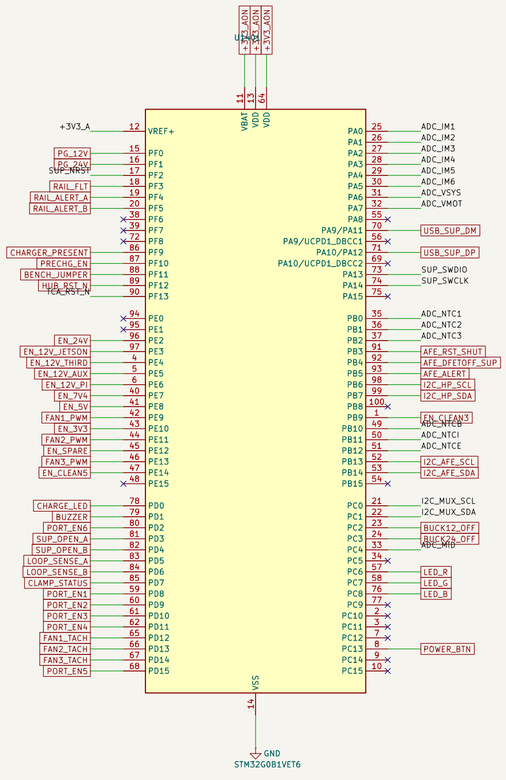


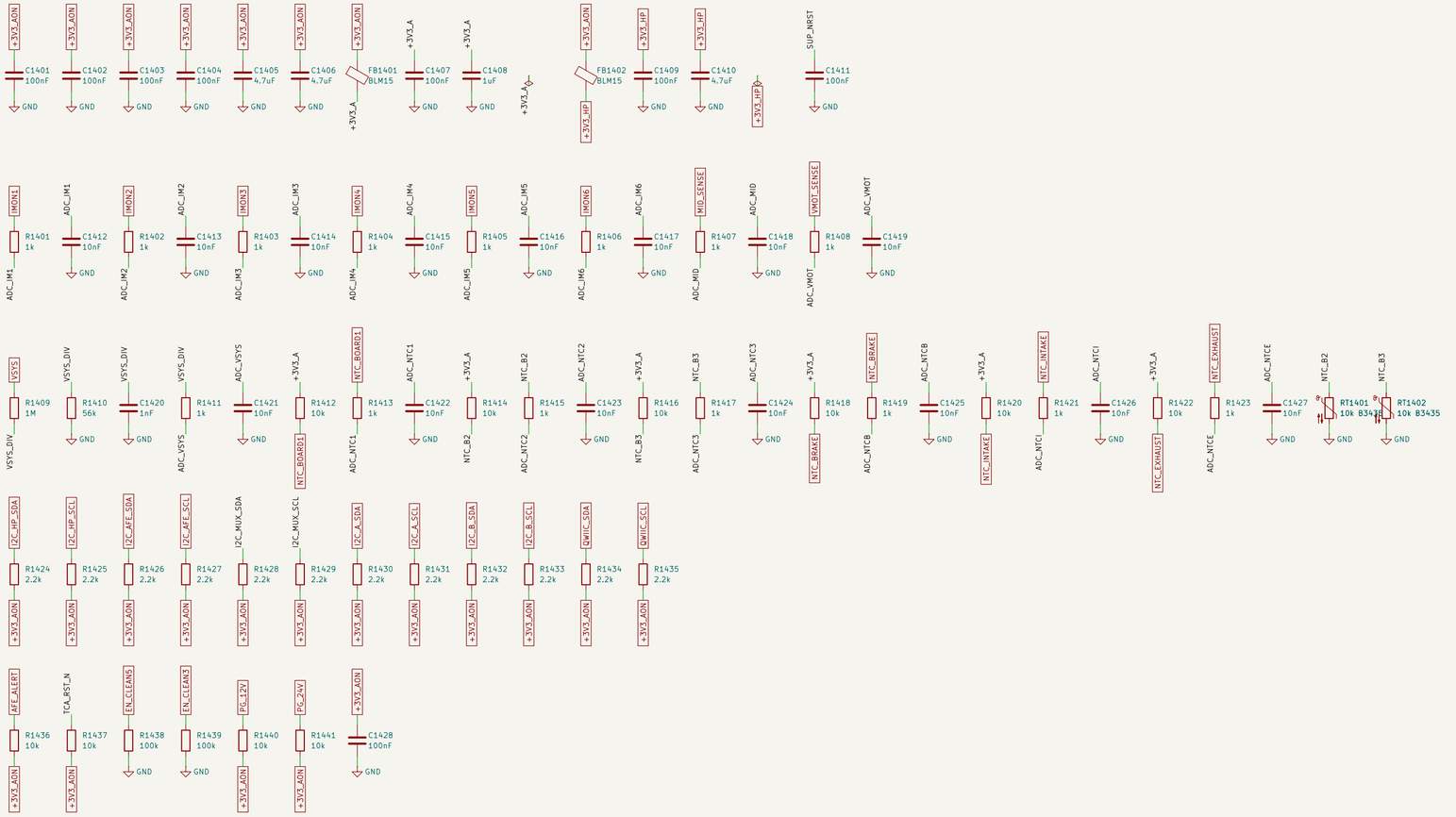
marengo

Sheet: /LP_logic_board/
File: LP_logic_board.kicad_sch

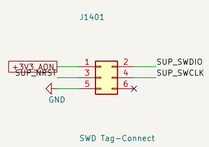
Title: LP_logic_board: low-power board

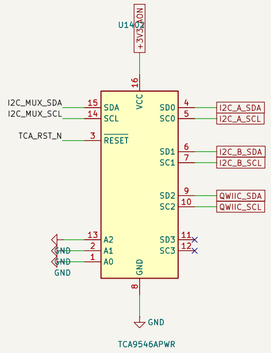
Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 13/27





G0B1VET6 pin plan: parts check 68 of 68 against ST pin data. Extra lines beyond the plan: PC2/PC3 buck off, PF0/PF1 PG, PF3–5 rail flags, PB9 EN_CLEAN3. PF10 was BOARD_PRESENT, now PRECHG_EN. PC5 free.
I2C: 3 hardware buses. I2C1 HP meters. I2C2 AFE. I2C3 to TCA9546A at 0x70 (ch0 meters A, ch1 meters B, ch2 Qwiic, ch3 spare).
Five NTC channels sit on 10k to the ADC reference. NTC_BOARD1 also reads open when the HP board is missing: replaces BOARD_PRESENT.





Supervisor can open the banks and ports, never close them against the loop. All enables fail off.

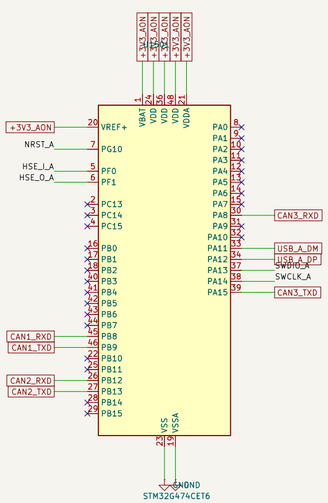
marengo

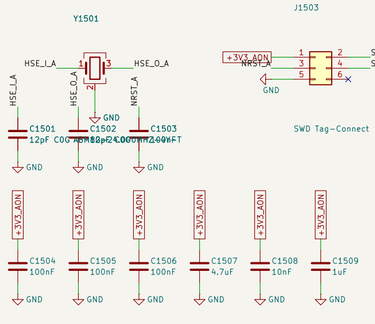
Sheet: /LP_logic_board/supervisor_mcu/
File: supervisor_mcu.kicad_sch

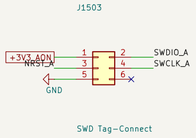
Title: supervisor_mcu: STM32G0B1 supervisor, I2C mux

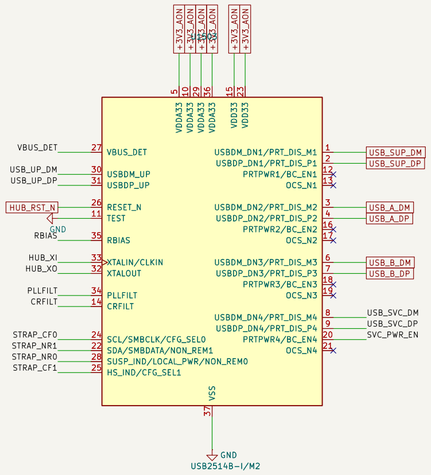
Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 14/27

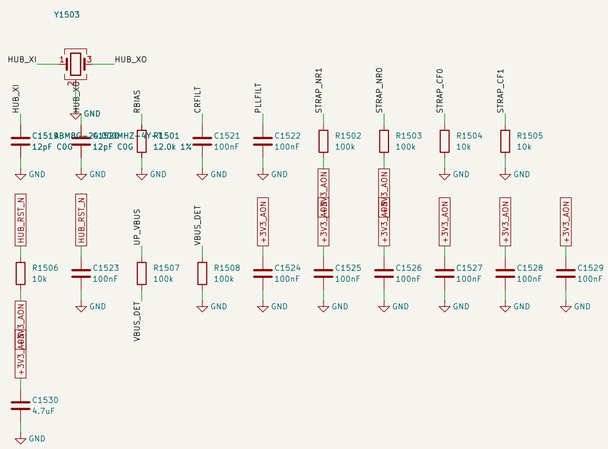
Hub: DN1 supervisor, DN2 bridge A, DN3 bridge B, DN4 service. No EEPROM: CFG_SEL 00, NON_REM 11, self powered, individual switching. TEST to GND. OCS_N1–4 open (internal pull-up).
FDCAN1_RX on PB8-BOOT0: program option bytes to ignore the pin before first run. PA15 released from JTDI for FDCAN3_TX. Firmware: budgetcan_fw, candleLight has no G474 build.
Service USB-C sits on a hub downstream port, so it is a host port: a laptop cannot reach the board through it. Rp 56k, default USB current.

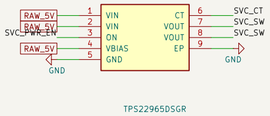


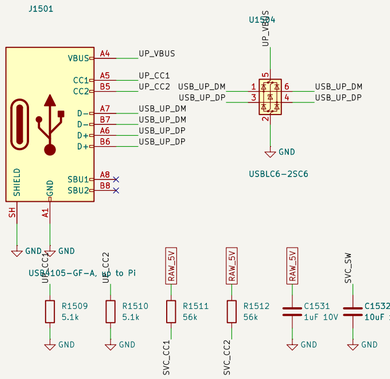


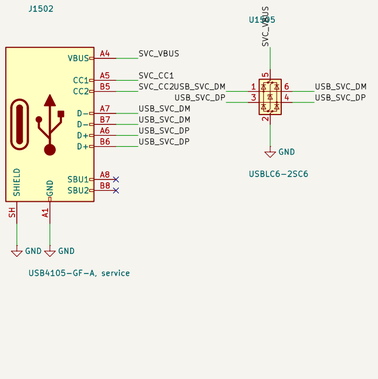












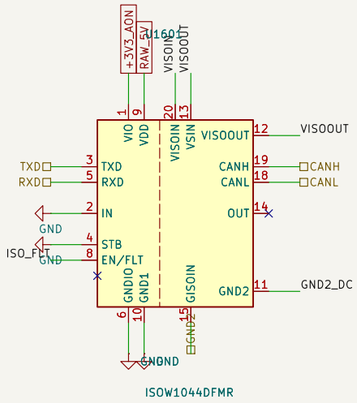
6 CAN buses over one USB-C to the Pi. Hub comes up without the supervisor, so the Pi keeps CAN if

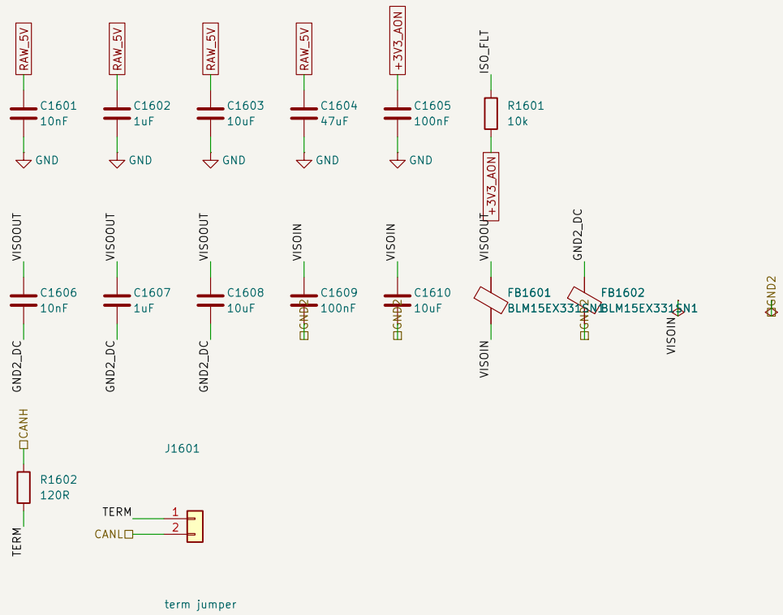
marengo

Sheet: /LP_logic_board/can_bridges_usb_hub/
File: can_bridges_usb_hub.kicad_sch

Title: can_bridges_usb_hub: two STM32G474 CAN bridges, USB2514B hub

Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 15/27





VDD 5 V is required: 6 ports draw up to 0.74 A from RAW_5V. Bus pins +/−58 V abs, +/−45 V differential. STB tied to GNDIO for normal mode. IN grounded, OUT open: spare isolated GPIO.
EN/FLT 10k pull−up: enabled, fault readable with a scope. Termination 120 Ohm by jumper per port, far end at the trunk.

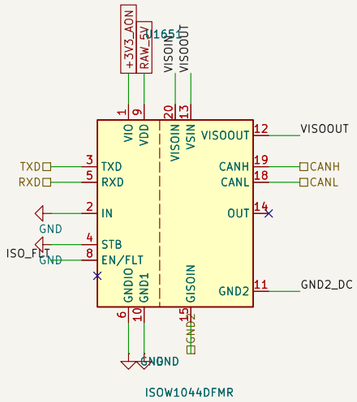
ISOW1044: integrated isolated supply. 4 mm metal keep−out around pins 11 and 12.

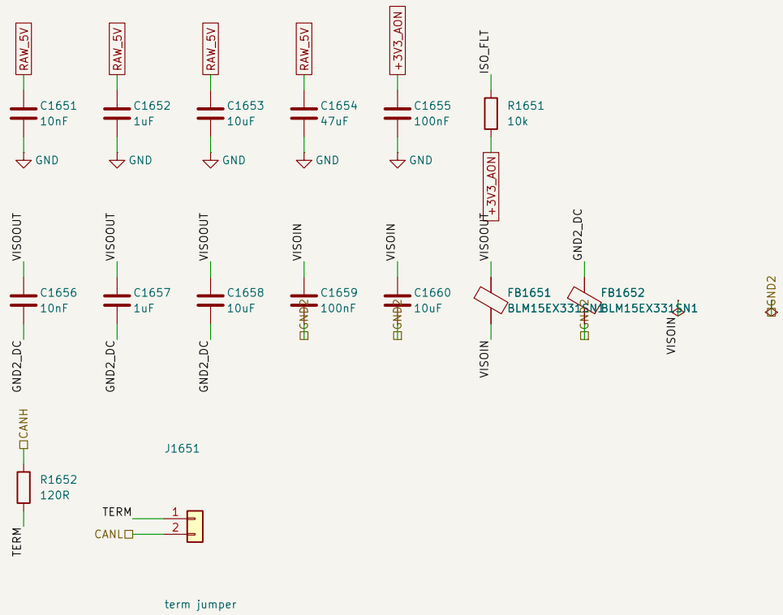
marengo

Sheet: /LP_logic_board/isolated_can_1/
File: isolated_can.kicad_sch

Title: isolated_can: one isolated CAN channel (instanced x6)

Size: A3	Date: 2026−11−06	**Rev: A−wip**
KiCad E.D.A. 10.0.8		Id: 16/27





VDD 5 V is required: 6 ports draw up to 0.74 A from RAW_5V. Bus pins +/−58 V abs, +/−45 V differential. STB tied to GNDIO for normal mode. IN grounded, OUT open: spare isolated GPIO.
EN/FLT 10k pull-up: enabled, fault readable with a scope. Termination 120 Ohm by jumper per port, far end at the trunk.

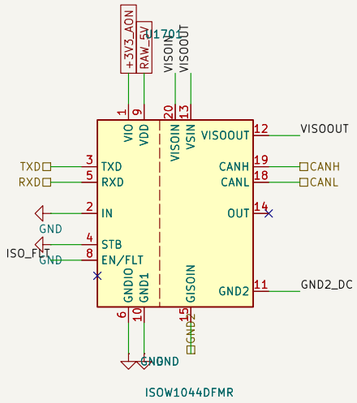
ISOW1044: integrated isolated supply. 4 mm metal keep-out around pins 11 and 12.

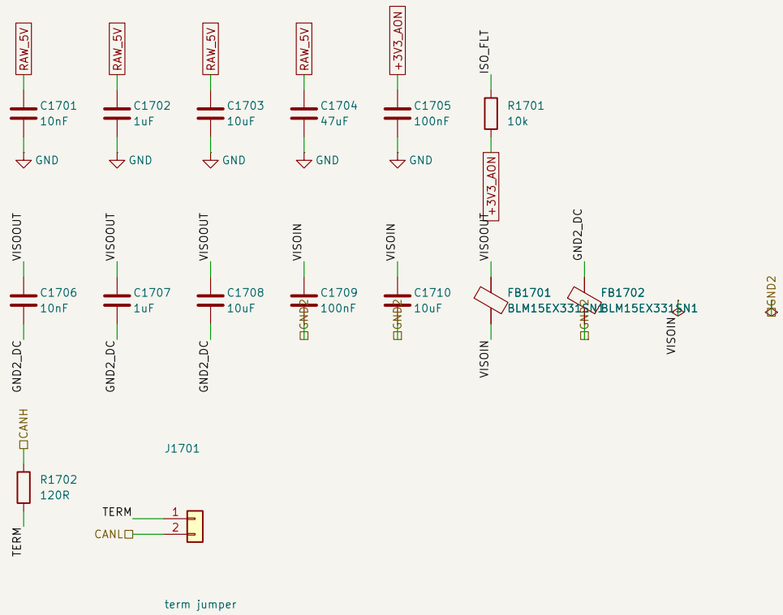
marengo

Sheet: /LP_logic_board/isolated_can_2/
File: isolated_can.kicad_sch

Title: isolated_can: one isolated CAN channel (instanced x6)

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 17/27





VDD 5 V is required: 6 ports draw up to 0.74 A from RAW_5V. Bus pins +/−58 V abs, +/−45 V differential. STB tied to GNDIO for normal mode. IN grounded, OUT open: spare isolated GPIO.
EN/FLT 10k pull-up: enabled, fault readable with a scope. Termination 120 Ohm by jumper per port, far end at the trunk.

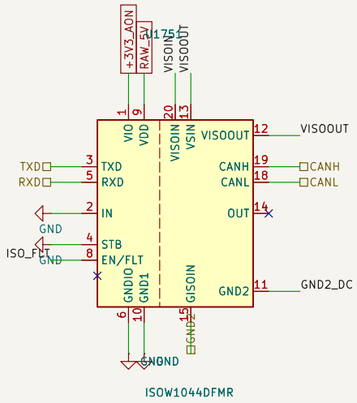
ISOW1044: integrated isolated supply. 4 mm metal keep-out around pins 11 and 12.

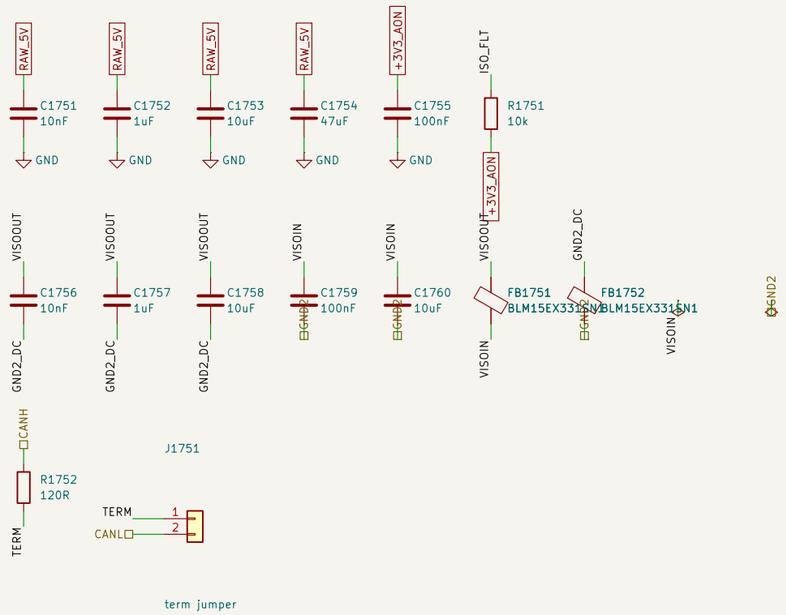
marengo

Sheet: /LP_logic_board/isolated_can_3/
File: isolated_can.kicad_sch

Title: isolated_can: one isolated CAN channel (instanced x6)

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 18/27





VDD 5 V is required: 6 ports draw up to 0.74 A from RAW_5V. Bus pins +/−58 V abs, +/−45 V differential. STB tied to GNDIO for normal mode. IN grounded, OUT open: spare isolated GPIO.
EN/FLT 10k pull−up: enabled, fault readable with a scope. Termination 120 Ohm by jumper per port, far end at the trunk.

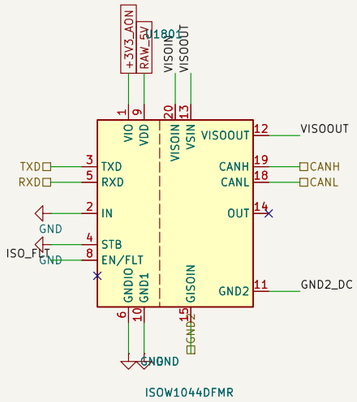
ISOW1044: integrated isolated supply. 4 mm metal keep−out around pins 11 and 12.

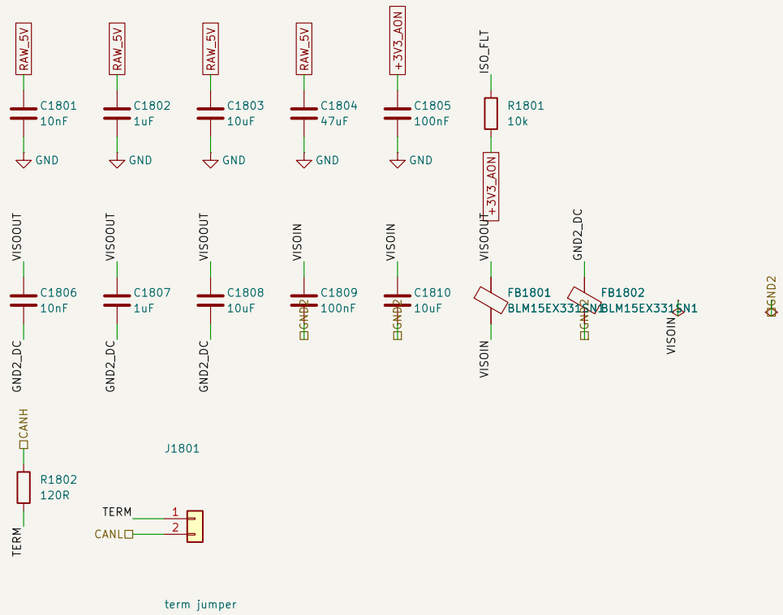
marengo

Sheet: /LP_logic_board/isolated_can_4/
File: isolated_can.kicad_sch

Title: isolated_can: one isolated CAN channel (instanced x6)

Size: A3	Date: 2026−11−06	**Rev: A−wip**
KiCad E.D.A. 10.0.8		Id: 19/27





VDD 5 V is required: 6 ports draw up to 0.74 A from RAW_5V. Bus pins +/−58 V abs, +/−45 V differential. STB tied to GNDIO for normal mode. IN grounded, OUT open: spare isolated GPIO.
EN/FLT 10k pull-up: enabled, fault readable with a scope. Termination 120 Ohm by jumper per port, far end at the trunk.

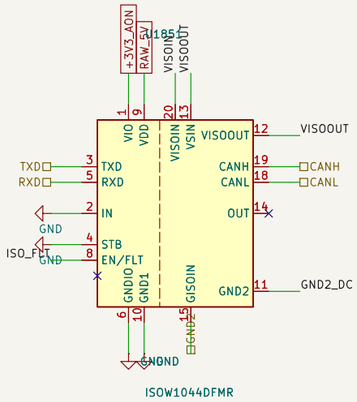
ISOW1044: integrated isolated supply. 4 mm metal keep-out around pins 11 and 12.

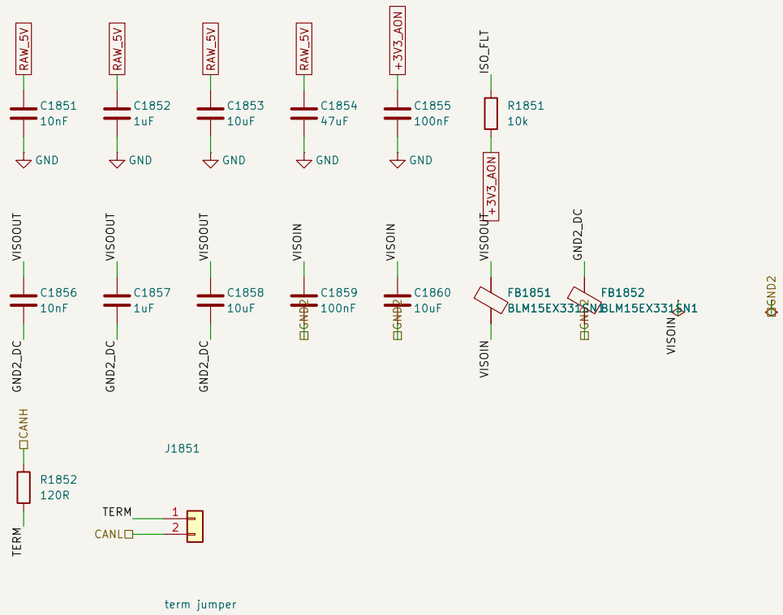
marengo

Sheet: /LP_logic_board/isolated_can_5/
File: isolated_can.kicad_sch

Title: isolated_can: one isolated CAN channel (instanced x6)

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 20/27





VDD 5 V is required: 6 ports draw up to 0.74 A from RAW_5V. Bus pins +/−58 V abs, +/−45 V differential. STB tied to GNDIO for normal mode. IN grounded, OUT open: spare isolated GPIO.
EN/FLT 10k pull-up: enabled, fault readable with a scope. Termination 120 Ohm by jumper per port, far end at the trunk.

ISOW1044: integrated isolated supply. 4 mm metal keep-out around pins 11 and 12.

marengo

Sheet: /LP_logic_board/isolated_can_6/		
File: isolated_can.kicad_sch		
Title: isolated_can: one isolated CAN channel (instanced x6)		
Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 21/27

Ideal diode and hold-up. LM5050-1 not LM74700: PLAN 2.3 wants 75 V; LM74700 stops at 65 V, and VSYS needs no reverse-polarity function here.
1.32 mF: 12 ms at 90 W from 46.8 V, 6.6 ms from 37 V. README claimed 1 mF for 10 ms: true only above 48 V.
The 24 V converter taps VSYS ahead of the diode so 75 W does not drain the hold-up.

U1101
OR_GATE
VSYS
VSYS_OR
GND
LM5050MK-1
Q1101
CSD19532Q5B
U1102
EN_12
RT_12
SS_12
FB_12
VCC_12
ILIM_12
VSYS_OR
COMP_12
SYNC_24
PG_12V
LO_12
VCC_12
BST_12
HO_12
SW12
Q1102
Q1103
CSD19534Q5A
BSC040N10NS5ATMA1
LO_12
SW12
LM5145RGYR
Q1104
BUCK12_OFF
BSS123
+3V3_AON
R1111
100k
BUCK12_OFF
L1101
XAL1510-682MED 33.2k
R1101
R1102
33.2k
R1103
2.37k
R1104
6.04k
C1107
10nF
C1108
33pF
R1105
100R
C1109
1nF
C1110
33nF
C1111
100nF 25V
R1106
2.2R
C1112
2.2uF 25V
R1107
348R
C1113
18pF C0G 100V
R1108
1.21M
R1109
49.9k
C1114
100nF 100V
R1110
20k
C1101
100nF 100V
C1102
100nF 100V
C1103
EEV-FK1K331M 330uF 80V
C1104
C1105
C1106
+12V
+24V
+3V3_AON
C1115
4.7uF 100V
C1116
4.7uF 100V
C1117
4.7uF 100V
C1118
4.7uF 100V
C1119
4.7uF 100V
C1120
22uF 25V
C1121
22uF 25V
C1122
22uF 25V
C1123
22uF 25V
C1124
22uF 25V
C1125
22uF 25V
C1126
22uF 25V

U1103
EN_24
RT_24
SS_24
FB_24
SYNC_24
ILIM_24
VSYS
COMP_24
HO_24
PG_24V
LO_24
VCC_24
BST_24
SW24
Q1105
CSD19534Q5A
LM5145RGYR
Q1107
BUCK24_OFF
BSS123
R1122
100k
BUCK24_OFF
L1102
XAL1510-333MED 33.2k
R1112
R1113
80.6k
R1114
2.8k
R1115
17.8k
C1127
6.8nF
C1128
10pF
R1116
100R
C1129
680pF
C1130
33nF
C1131
100nF 25V
R1117
2.2R
C1132
2.2uF 25V
R1118
301R
C1133
22pF C0G 100V
R1119
499k
R1120
18.2k
C1134
100nF 100V
R1121
20k
C1135
4.7uF 100V
C1136
4.7uF 100V
C1137
4.7uF 100V
C1138
10uF 50V
C1139
10uF 50V
C1140
10uF 50V
C1141
10uF 50V
C1142
10uF 50V
C1143
10uF 50V
C1144
10uF 50V

BUCK12_OFF, BUCK24_OFF: pulled up by the always-on rail so a sleeping or reset supervisor holds both converters off. Supervisor drives low to start them (Standby state, PLAN 3.2).
The 12 V converter is the 300 kHz master; SYNCOUT drives the 24 V converter 180 degrees out of phase. 12 V feeds its own VCC through a Schottky after start-up (lower loss).

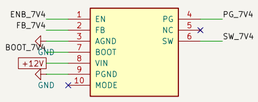
3.3 V always-on. EN tied to VIN: on whenever VSYS exists. Ship mode removes VSYS. 400 kHz, 3.31 V.

U1104
VSYS
EN_AON
RT_AON
FB_AON
PG_AON
BOOT_AON
SW_AON
LMR38010SDDAR
L1103
XAL6060-223MEC 22uH
R1123
R1124
43.2k
R1125
64.9k
C1145
4.7uF 100V
C1146
100nF 100V
C1147
100nF
C1148
22uF 25V
C1149
22uF 25V
C1150
22uF 25V
R1126
100k

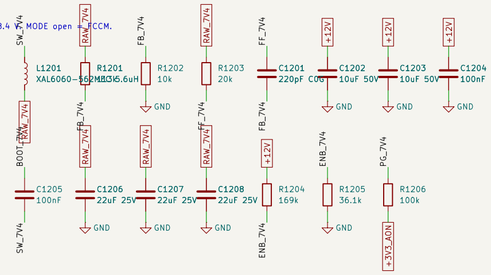
LM5050-1, LM5145 x2, LMR38010. Hand design, values in RATIONALE-log.
marengo
Sheet: /LP_logic_board/front_end_12v_24v/
File: front_end_12v_24v.kicad_sch
Title: front_end_12v_24v: ideal diode, hold-up, 12 V, 24 V, 3.3 V always-on
Size: A2 | Date: 2026-11-06 | Rev: A-wip
KiCad E.D.A. 10.0.8 | Id: 22/27

Servo rail. Adjust by R top: 90.9k = 6.05 V, 113k = 7.4 V, 130k = 8.4 V. MODE open = FCCM.

U1201

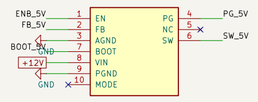


TPS56637RPAR

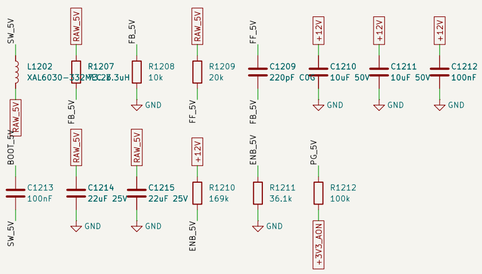


5 V peripheral, 4 A. EN divider per TI: starts near 10 V input.

U1202

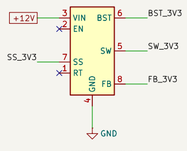


TPS56637RPAR

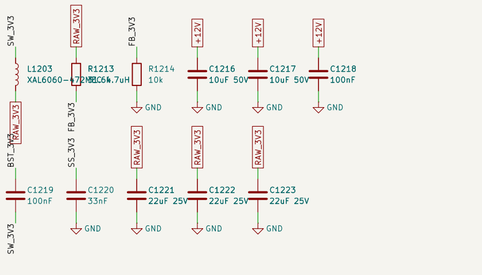


3.3 V peripheral, 2 A. RT open = 500 kHz, EN open = on.

U1203

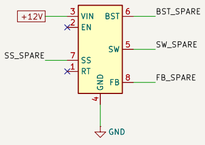


TPS62933DRLR

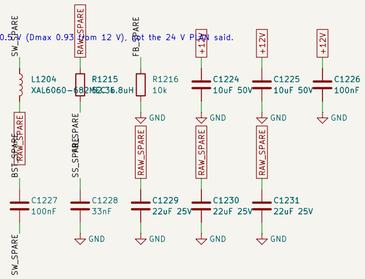


Spare rail, stuffed 5 V. 3.3 V: R top 31.6k. 9 V: see log. Max about 10.5 V (Dmax 0.93 from 12 V), but the 24 V PSU said.

U1204

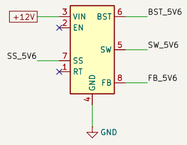


TPS62933DRLR

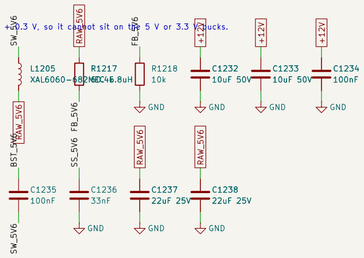


New: 5.6 V pre-regulator. TPS7A20 needs VIN <= 6.0 V and >= VOUT +0.3 V, so it cannot sit on the 5 V or 3.3 V bucks.

U1205

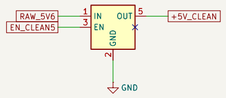


TPS62933FDRLR

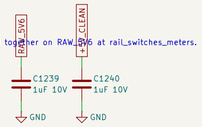


Clean rails, 0.3 A each, under 5 mV ripple. Metered together on RAW_5V6 at rail_switches_meters.

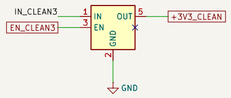
U1206



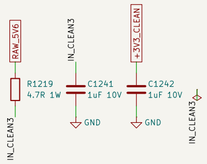
TPS7A2050PDBVR



U1207



TPS7A2033PDBVR





TPS56637 x2, TPS62933 x3, TPS7A20 x2. Hand design, values in RATIONALE-log.

marengo

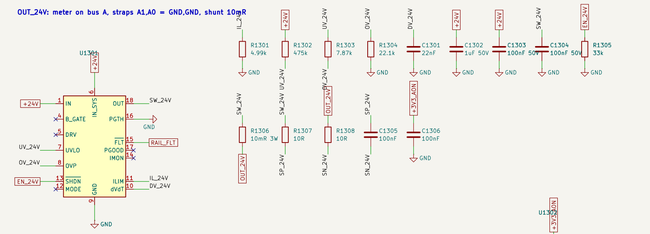
Sheet: /LP_logic_board/pol_rails/

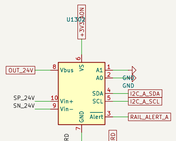
File: pol_rails.kicad_sch

Title: pol_rails: point-of-load rails from 12 V

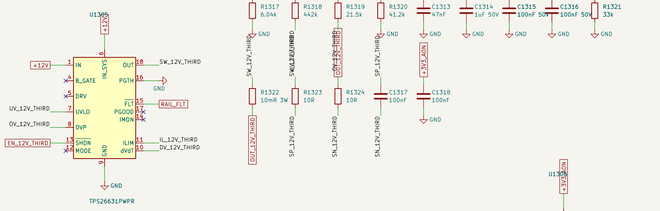
Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 23/27

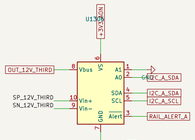
OUT_24V: meter on bus A, straps A1,A0 = GND,GND, shunt 10mR



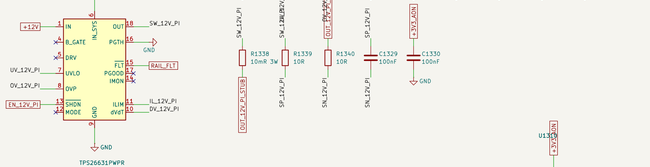


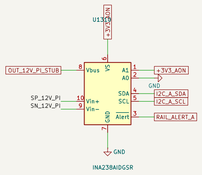
OUT_12V_THIRD: meter on bus A, straps A1,A0 = GND,SDA, shunt 10mR



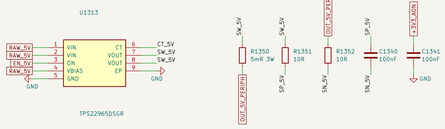


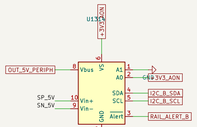
OUT_12V_PI_STUB: meter on bus A, straps A1,A0 = VS,GND, shunt 10mR



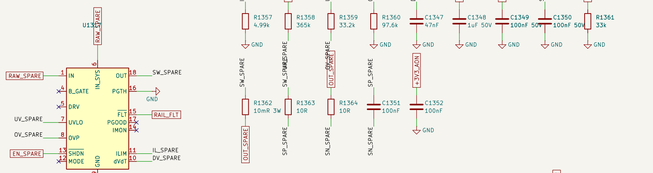


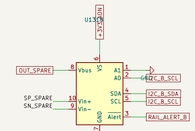
OUT_5V_PERIPH: meter on bus B, straps A1,A0 = GND,VS, shunt 5mR





OUT_SPARE: meter on bus B, straps A1,A0 = GND,SCL, shunt 10mR

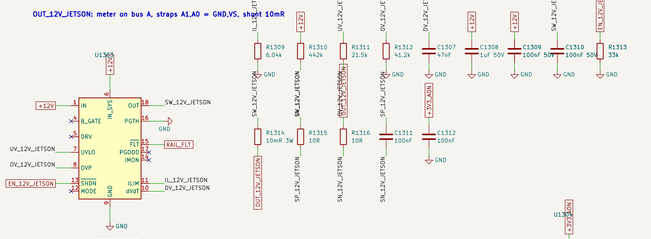


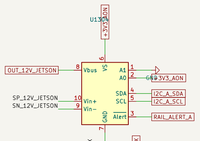


TPS26631 UV/OV chain mandatory (UVLO to GND means 15.46 V, would kill a 12 V rail). Latch mode, supervisor clears. SHDN default off.
TPS22965 has no current limit and its 6 A is under the buck limit; output polyfuse still owed (PLAN asks for a limit).
INA238 has no energy registers; firmware integrates at 100 Hz, or swap to INA228 (same footprint).

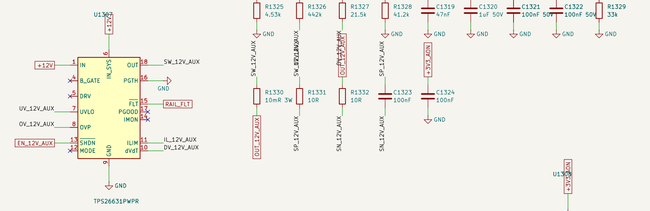
R1368 100k
R1369 10k
R1370 10k

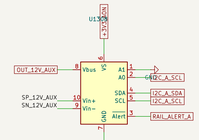
OUT_12V_JETSON: meter on bus A, straps A1,A0 = GND,VS, shunt 10mR



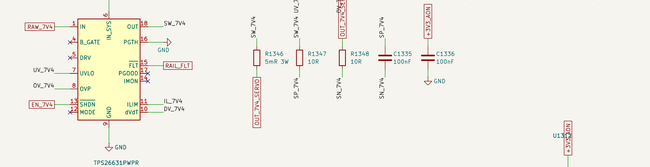


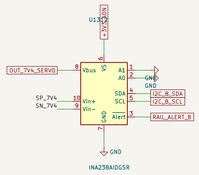
OUT_12V_AUX: meter on bus A, straps A1,A0 = GND,SCL, shunt 10mR



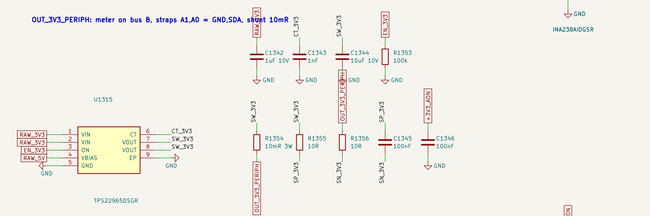


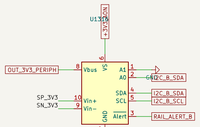
OUT_7V4_SERVO: meter on bus B, straps A1,A0 = GND,GND, shunt 5mR



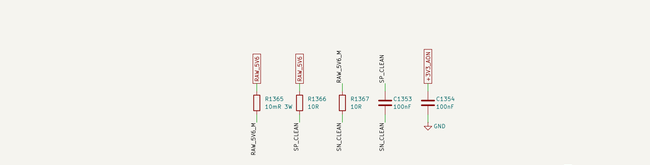


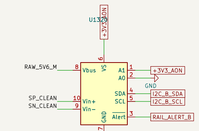
OUT_3V3_PERIPH: meter on bus B, straps A1,A0 = GND,SDA, shunt 10mR





RAW_5V6_M: meter on bus B, straps A1,A0 = VS,GND, shunt 10mR





TPS26631 eFuse / TPS22965 switch + INA238 per output. 2 I2C buses, 5 meters each.

marengo

Sheet: /LP_logic_board/rail_switches_meters/

File: rail_switches_meters.kicad_sch

Title: rail_switches_meters: rail switches and meters

Size: A1	Date: 2026-11-06	Rev: A-wip
KiCad E.D.A. 10.0.8		Id: 24/27

E-stop loop, 6.9 mA: +12V, 680R, NC contacts, opto A, opto B, 620R, GND. Open loop, lost 12 V or lost 3V3 all give EN low.

J2001

J2002

J2003

tether, shorting dummy

wireless, shorting plug

E-stop loop

U2001

ISOM8710

U2003

ISOM8710

U2002

SN74LVC1G14DBV

U2004

SN74LVC1G14DBV

Q2001

2N7002

Q2002

2N7002

R2001 680R 0.25W

R2002 620R 0.25W

D2001 1N4148W

D2002 SMAJ15A

D2003 SMAJ15A

C2001 100nF

C2002 100nF

C2003 100nF

C2004 100nF

R2003 0R Cat 0 link

R2004 1k

R2005 100k

R2006 1k

R2007 160k

C2005 2.2uF

R2008 1M

D2004 BAT54

R2009 0R Cat 0 link

R2010 1k

R2011 100k

R2012 1k

R2013 160k

C2006 2.2uF

R2014 1M

D2005 BAT54

SUP_OPEN high: 2N7002 sinks EN. MCU dead or Hi-Z: pull-up opens the banks. The FET only sinks, so firmware cannot close an open loop.
Cat 1 option: remove the 0R, fit the four DNP parts. 289 ms nominal (232–356 ms). DNP in rev A: Cat 0.
Not Cat 3: one series loop feeds both channels, a jumping short across it defeats both.

Fans, 12 V, 25 kHz PWM open drain (polarity inverts in firmware).

J2004

J2005

J2006

fan 1

fan 2

fan 3

Q2003

2N7002

Q2004

2N7002

Q2005

2N7002

R2015 10k

R2016 10k

R2017 10k

R2018 100k

R2019 100k

R2020 100k

User I/O: 6-pin shell cable, the plan's 4 pins cannot carry RGB + button. Bench jumper, buzzer, charge LED, Qwiic, air NTC.

J2007

J2008

J2009

J2010

J2011

J2012

shell: button + RGB

charge LED

bench jumper

Qwiic

intake NTC

exhaust NTC

R2021 330R

R2022 330R

R2023 330R

R2024 330R

R2025 10k

R2026 10k

D2006 BAT54

R2027 100k

TP2001

TP2002

D2007 1N4148W

Q2006

2N7002

LS2001

PS1240P02BT

Rail outputs, one Micro-Fit each. TS2 wake: button low pulls AFE TS2 to VSS through 10k + BAT54 with the board off.

J2013

J2014

J2015

J2016

J2017

J2018

J2019

J2020

J2021

J2022

J2023

OUT_24V

OUT_12V_JETSON

OUT_12V_THIRD

OUT_12V_AUX

OUT_12V_PI_STUB

OUT_7V4_SERVO

OUT_5V_PERIPH

OUT_3V3_PERIPH

OUT_SPARE

5V_CLEAN

3V3_CLEAN

E-stop path: loop current to opto to inverter to gate-driver enable. No firmware in that path.

marengo

Sheet: /LP_logic_board/io_fans_estop_loop/

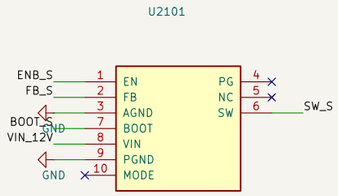
File: io_fans_estop_loop.kicad_sch

Title: io_fans_estop_loop: E-stop loop, fans, user I/O, output connectors

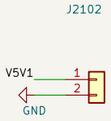
Size: A2	Date: 2026-11-06	Rev: A-wip
KiCad E.D.A. 10.0.8		Id: 25/27



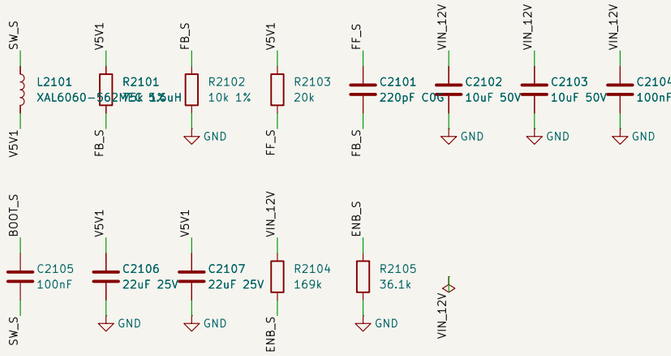
12 V in



TPS56637RPAR



5.1 V out, 8.5 A per contact



Output pigtail goes to a USB-C plug. A 2x3 housing on the GPIO header would put all 6 A through one GND pin (pin 6) and cannot lock.
Set usb_max_current_enable=1 on the Pi. PG left open: no logic rail on the stub.

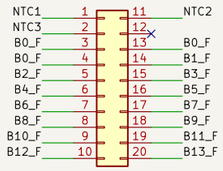
Own 2-layer board. Local nets; harness map checked in check.py.

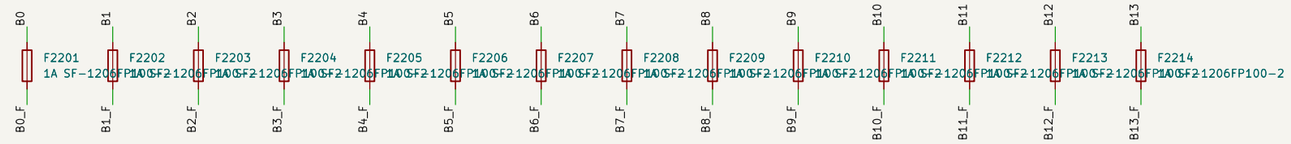
marengo

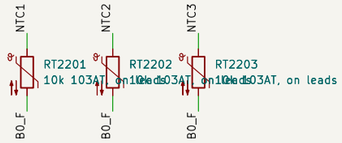
Sheet: /pi_power_stub/
File: pi_power_stub.kicad_sch

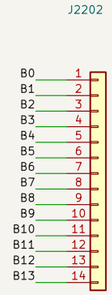
Title: pi_power_stub: 12 V to 5.1 V 6 A at the Pi

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 26/27









Fuse at the cell, one per tap. Presence: pin 13 tied to the B0 return. 1 A tap: 0.1455 Ohm, 63 V, 50 A interrupt (below the pack short-circuit current: rely on wire impedance or a higher breaking part, open).
Balance current through the fuse adds about 44 mV of sense error at 150 mA across two taps.

Own 2-layer board at the pack. Pin map must match HP J2. Local nets; harness map checked in che

marengo

Sheet: /pack_sense_board/
File: pack_sense_board.kicad_sch

Title: pack_sense_board: tap fuses, NTC, presence loop

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 27/27